

Model	Dataset	Split Method	Metric	 Drug & Cell Line Features	 One-Hot Encoding	 Percent Deviation (%)
DeepSynergy	O'Neil	LPO	MSE	252.55 ± 12.46	250.23 ± 13.24	-0.92 ± 1.76
			PCC	0.72	0.73	1.39
			SCC	0.72	0.73	1.39
DeepDDS	O'Neil	LTO	ROC AUC	0.93	0.94	1.08
			PR AUC	0.93	0.94	1.08
			ACC	0.85	0.86	1.18
Matchmaker	DrugComb	LPO	MSE	97.23 ±1.14	100.25 ±1.15	3.11 ±0.54
			PCC	0.76	0.75	-1.32
			SCC	0.72	0.70	-2.78
MARSY	DrugComb	LPO	MSE	32.94 ± 1.20	30.26 ± 1.11	-8.14 ± 1.49
			PCC	0.87	0.88	1.15
			SCC	0.74	0.75	1.35
JointSyn	O'Neil	LTO	MSE	76.56 ±0.48*	77.38 ±2.21	1.07
			PCC	0.89	0.88	-1.12
			R ²	0.78	0.78	0.00